35021P28	R PROGRAMMING	Category	L	T	P	Credit
		EC-PS	3	0	0	3

PREAMBLE:
R is a scripting language for statistical data manipulation and analysis. It was inspired by and is most compatible with the statistical language.

PREREQUISITE:
NIL

COURSE OBJECTIVES

1	To learn R Programming
2	To Study Object Oriented Programming
3	To Study Functional Programming

COURSE OUTCOMES

On the successful completion of the course, students will be able to

CO1: To Understand the basics in R programming in terms of constructs, control statements, string functions	Understand & Apply
CO2: To Understand the use of R for Big Data analytics	Understand & Apply
CO3: Learn to apply R programming for Text processing	Understand & Apply
CO4: Able to appreciate and apply the R programming from a statistical perspective	Understand & Apply
CO5: To learn Big Data	Understand & Apply

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

COS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO 3
CO1	S	S	M	M	L	S	S	M	S	L	S	-	-	-	-
CO2	M	S	M	M	M	S	S	M	S	M	M	-	-	-	-
CO3	S	S	S	S	M	S	S	S	S	M	S	-	-	-	-
CO4	S	S	S	M	M	S	S	S	S	L	S	-	-	-	-
CO5	S	S	M	M	L	S	M	M	S	L	M	-	-	-	-

S- Strong; M-Medium; L-Low

SYLLABUS

INTRODUCTION

Introducing to R – R Data Structures – Help functions in R – Vectors – Scalars – Declarations – recycling – Common Vector operations – Using all and any – Vectorized operations – NA and NULL values – Filtering – Vectorised if-then else – Vector Equality – Vector Element names

MATRICES, ARRAYS AND LISTS

Creating matrices – Matrix operations – Applying Functions to Matrix Rows and Columns – Adding and deleting rows

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and columns – Vector/Matrix Distinction – Avoiding Dimension Reduction – Higher Dimensional arrays – lists – Creating lists – General list operations – Accessing list components and values – applying functions to lists – recursive lists

DATA FRAMES

Creating Data Frames – Matrix-like operations in frames – Merging Data Frames – Applying functions to Data frames – Factors and Tables – factors and levels – Common functions used with factors – Working with tables - Other factors and table related functions - Control statements – Arithmetic and Boolean operators and values – Default values for arguments - Returning Boolean values – functions are objects – Environment and Scope issues – Writing Upstairs - Recursion – Replacement functions – Tools for composing function code – Math and Simulations in R

OOP

S3 Classes – S4 Classes – Managing your objects – Input/Output – accessing keyboard and monitor – reading and writing files – accessing the internet – String Manipulation – Graphics – Creating Graphs – Customizing Graphs – Saving graphs to files – Creating three-dimensional plots

INTERFACING

Interfacing R to other languages – Parallel R – Basic Statistics – Linear Model – Generalized Linear models – Non-linear models – Time Series and Auto-correlation – Clustering

TEXT BOOKS:

3. The Art of R Programming: A Tour of Statistical Software Design, Norman Matloff, No Starch Press, 2011
4. R for Everyone: Advanced Analytics and Graphics, Jared P. Lander, Addison-Wesley Data & Analytics Series, 2013.

REFERENCES:

3. Beginning R – The Statistical Programming Language, Mark Gardener, Wiley, 2013.
4. Introductory R: A Beginner's Guide to Data Visualisation, Statistical Analysis and Programming in R, Robert Knell, Amazon Digital South Asia Services Inc, 2013

COURSE DESIGNERS

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